

Xiaoxuan Zhou

151-5201-2670 ◇ echoaugust2@gmail.com

EDUCATION

Software College, Northeastern University (China)
B.E. in Software Engineering

September 2022 – June 2026

School of Computer Science, Peking University
Incoming Ph.D. Student (Five-year Direct Ph.D. Program)

September 2026 – Expected

ACADEMIC PERFORMANCE

Academic Year	Ranking
2022-2026	1st in GPA (1/297)
2022-2025	1st in Comprehensive Quality (1/496)

ENGLISH PROFICIENCY

IELTS	7.0	September 2022
CET-6	587	December 2024

RESEARCH

TestTailor: Generating High-Coverage Tests via Path-Proximal Tests with LLMs

Xiaoxuan Zhou, Yiling Lou, Jinhao Dong, Dan Hao

FSE 2026

- **Advisor:** Dan Hao, School of Computer Science, Peking University.
- **Description:** TestTailor transforms vague coverage goals into precise path-level guidance for LLM-based test generation by combining path-proximal test analysis with symbolic constraints from uncovered paths, explicitly telling the model how to reach hard-to-cover branches. On the CODAMOSA benchmark, it significantly outperforms prior methods, achieving up to 12.78% higher coverage while reducing API cost by about 60%.

TestDecision: Sequential Test Suite Generation via Greedy Optimization and Reinforcement Learning

Guoqing Wang, Chengran Yang, Xiaoxuan Zhou, Zeyu Sun, Bo Wang, David Lo, Dan Hao *ISSTA 2026*

- **Advisor:** Dan Hao, School of Computer Science, Peking University.
- **Description:** TestDecision introduces a step-wise greedy inference framework theoretically justified under monotone submodularity, paired with a reinforcement learning training pipeline to maximize marginal coverage gain. It achieves 38.15–52.37% improvement in branch coverage and 298.22–558.88% improvement in execution pass rate over base models, finding 58.43–95.45% more bugs than vanilla LLMs.

Code-Vision: Evaluating Multimodal LLMs Logic Understanding and Reasoning Capabilities Through Code Generation

ICASSP 2026

- **Abstract:** We construct Code-Vision, a novel benchmark for evaluating the logical understanding and reasoning capabilities of MLLMs in code generation tasks. Code-Vision uses flowcharts as the primary input, making visual comprehension essential for generating correct code.

Boosting Text-To-Image Generation via Multilingual Prompting in Large Multimodal Models

Yongyu Mu, Hengyu Li, Junxin Wang, Xiaoxuan Zhou, Chenglong Wang, Yingfeng Luo, Qiaozhi He, Tong Xiao, Guocheng Chen and Jingbo Zhu *ICASSP 2025*

- **Advisor:** Tong Xiao, Director of Natural Language Processing Lab, Northeastern University.

- **Abstract:** We introduce PMT2I, a novel Parallel Multilingual Prompting method for Text-to-Image (T2I) tasks, which enhances the performance of MLLMs by leveraging multilingualism.

PROJECT

AI Agent Project

May 2024 - December 2024

- **Advisor:** Ning Li, School of Economics and Management, Tsinghua University.
- **Description:** We use multi-agent frameworks such as AutoGen and Langchain to develop a series of multi-agent applications. These include paper writing with multi-agent collaboration, agent-based negotiations, and a project focused on guiding learners through conversational learning.

HONORS AND AWARDS

Top 10 Outstanding Undergraduate Students of Northeastern University

April 2026

Outstanding Graduate of Liaoning Province

December 2025

National Scholarship (Twice)

December 2023, November 2024

Outstanding College Students of Shenyang City (Only One)

December 2023